

Yuan Li

Phoenix, AZ • 480-828-6987 • liyuan6006@gmail.com
LinkedIn: <https://www.linkedin.com/in/yuan-li-b89b4974> • GitHub:
<https://github.com/liyuan6006>

Professional Summary

Lead Software Engineer / Architect with 20 years of hands-on experience designing, building, and modernizing enterprise-level applications and platforms. Expertise in full-stack development, cloud-native architecture, distributed systems, microservices, event-driven architecture, Agentic AI solutions, and machine learning.

Passionate about Agentic AI and machine learning, with hands-on experience building AI-powered applications and machine learning models. Solid knowledge of frameworks, technology stacks, and ecosystems in both AI and machine learning.

Proven leader in large-scale cloud migrations, platform modernization, database transformation, and high-availability system design with zero data loss and minimal downtime.

Experienced in leading engineering teams, collaborating with stakeholders, and driving end-to-end delivery from architecture and development through production deployment.

Technical Skills

- **Languages & Frameworks:** Python, C#, TypeScript, JavaScript, Go, .NET, NestJS, Angular, React, React Native, REST APIs, FastAPI.
- **AI & LLM:** Agentic AI Engineering, OpenAI API Integration, LLM Integration, RAG (Retrieval-Augmented Generation) Pipeline, Prompt Engineering, AI Evaluation & Validation, AI Guardrails, and AI Frameworks including LangChain, LlamaIndex, Semantic Kernel, LangGraph, LangSmith, Ragas, DeepEval, and Nemo Guardrails.
- **Machine Learning:** Feature Engineering, Model Building, Training, Evaluation, Hyperparameter Tuning, and Model Registration using tools/libraries such as PyTorch, TensorFlow, and MLflow.
- **Databases:** PostgreSQL, pgvector for RAG, SQL Server, MySQL, Oracle and NoSQL databases, such as MongoDB and Google Firestore.
- **Distributed Systems & Architecture:** Microservices, Distributed Systems Design, and Event-Driven Architecture using Apache Kafka, Spark, API Gateway, gRPC, and Go.
- **Observability:** OpenTelemetry and OpenSearch, Prometheus, Grafana.
- **Mobile Development:** Mobile Application Development for iOS & Android.
- **DevOps & Infrastructure:** CI/CD, Jenkins, Docker, Containerization, Kubernetes, Google Kubernetes Engine (GKE).
- **Cloud Platforms:** Microsoft Azure, Google Cloud Platform (GCP), AWS.

Personal Projects

<https://profile-agent.com/>

AI Resume Assistant Platform | RAG, OpenAI, LLMs, LangSmith. Built an AI-powered resume assistant using RAG, OpenAI, and LLMs that answers questions about a user's skills, experience, projects, and career history from uploaded profile data.

<https://yuanliproject.com/>

Banking Transaction Analytics Platform | React, ASP.NET Core, MongoDB, Kubernetes, Kafka. Developed a cloud-native transaction management platform with React and ASP.NET Core. Designed MongoDB data models and optimized large-scale transaction searches using indexes and Atlas Search. Deployed containerized services to Kubernetes on Google Cloud Platform.

Professional Experience

American Express – Lead Developer/Software Architect

Oct 2018 – Present

Technical Leadership & Architecture

- Led a team of 10 developers in a large-scale distributed systems migration, transforming 7 enterprise applications into independently deployable containerized microservices running in cloud environments.
- Directed a full database transformation from SQL Server to PostgreSQL, including schema, views, functions, constraints, triggers, and stored procedures. Planned and managed dependency sequencing to ensure a smooth and reliable migration process.
- Migrated the entire database from SQL Server to Postgres with zero data loss and no business disruption, maintaining full data integrity and relationships.
- Converted 7 legacy .NET Framework applications to .NET Core, enabling deployment on Linux environments without breaking existing functionality. Performed dependency analysis and planned the conversion strategy.
- Led modernization of enterprise frontend platforms by upgrading Angular applications from very old version to the latest version without breaking functionalities.
- Led technical problem-solving efforts by analyzing challenges, developing proofs of concept, and designing architectural solutions.
- Proposed new technologies, programming languages (such as React), and architectural approaches, and led their implementation.
- Oversaw end-to-end system architecture, ensuring scalability, reliability, and performance.
- Defined detailed project plans and timelines, communicated progress to stakeholders, and consistently delivered accurate estimates for a multi-year initiative.

Full-Stack Engineering & Solution Delivery

- Acted as a hands-on full-stack engineer, designing, developing, and delivering end-to-end enterprise solutions using a wide range of technologies and programming languages across front-end, APIs, and database platforms.
- Conducted research on new technologies and unfamiliar domains, delivering proof-of-concept solutions to guide direction.

Release & DevOps Management

- Implemented Git branching strategies to improve collaboration and release management.
- Managed the full release lifecycle from development through QA to production.

Cross-Team Collaboration

- Partnered with product owners to gather requirements and translate business needs into technical solutions.
- Defined features and created clear, executable user stories for the development team.
- Collaborated across multiple teams within the Amex ecosystem to align architecture, resolve dependencies, and drive project success.

Agile Leadership

- Served as Scrum Master, facilitating agile ceremonies, tracking sprint progress, and removing delivery blockers across cross-functional teams.
- Collaborated with product owners and stakeholders to refine requirements, estimate delivery timelines, and align engineering solutions with business objectives.

Infor – Senior Software Engineer

Oct 2014 – Oct 2018

- Contributed to the development and maintenance of large-scale CRM systems with millions of lines of code.
- Implemented new features while ensuring system stability and performance.
- Conducted code reviews and enforced engineering best practices.

CRM Factory (China) – Software Engineer

2010 – 2014

- Participated in migrating Porsche financial systems to a new platform.
- Supported system design, data migration, and implementation.

Education

- Master of Science in Engineering (Software Engineering) – Arizona State University
- Bachelor's Degree in International Economics – Renmin University of China